

AM-II Unit-I

Complex Analysis-I

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Complex Numbers and Their Geometric Representation

Definition: A number of the form $x + iy$, where x and y are real numbers and $i = \sqrt{-1}$, is called a complex number.

x is called the real part of $x + iy$ and y is called the imaginary part of $x + iy$. Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be two complex numbers.

Addition and Multiplication: The addition of z_1 and z_2 is defined by

$$z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2)$$

and multiplication of z_1 and z_2 is defined by

$$z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2) = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1).$$

Subtraction and Division: Subtraction and division are defined as the inverse operations of addition and multiplication, respectively. The difference $z_1 - z_2$ is given by

$$z_1 - z_2 = (x_1 + iy_1) - (x_2 + iy_2) = (x_1 - x_2) + i(y_1 - y_2)$$

and $\frac{z_1}{z_2}$ is given by

$$\frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2} = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i \frac{x_2y_1 - x_1y_2}{x_2^2 + y_2^2}.$$

Example

For $z_1 = 8 + 3i$ and $z_2 = 9 - 2i$, we get

$$z_1 - z_2 = (8 + 3i) - (9 - 2i) = -1 + 5i,$$

$$\frac{z_1}{z_2} = \frac{8 + 3i}{9 - 2i} = \frac{(8 + 3i)(9 + 2i)}{(9 - 2i)(9 + 2i)} = \frac{66 + 43i}{81 + 4} = \frac{66}{85} + i \frac{43}{85}.$$

Complex Conjugate Numbers The complex conjugate $\bar{z}$ of a complex number $z = x + iy$ is defined by $\bar{z} = x - iy$.

Polar Form of Complex Numbers. Powers and Roots

The complex plane becomes even more useful and gives further insight into the arithmetic operations for complex numbers if besides the xy -coordinates we also employ the usual polar coordinates r, θ defined by

$$x = r \cos \theta \quad \text{and} \quad y = r \sin \theta.$$

We see that then $z = x + iy$ takes the so-called *polar form*

$$z = r(\cos \theta + i \sin \theta).$$

r is called the *absolute value* or *modulus* of z and is denoted by $|z|$. Hence, $|z| = r = \sqrt{x^2 + y^2} = \sqrt{z\bar{z}}$. Geometrically, $|z|$ is the distance of the point z from the origin. Similarly, $|z_1 - z_2|$ is the distance between z_1 and z_2 .

θ is called the *argument* of z and is denoted by $\arg z$. Thus

$$\theta = \arg z = \arctan \frac{y}{x}.$$

Geometrically, θ is the directed angle from the positive x -axis. The principal value $\text{Arg } z$ of $\arg z$ is given by the double inequality

$$-\pi \leq \arg z \leq \pi.$$

Example

The complex number $z = 1 + i$ has the polar form

$z = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$. Therefore, $|z| = \sqrt{2}$, $\arg z = \frac{\pi}{4} \pm 2n\pi$
($n = 0, 1, 2, \dots$) and $\arg z = \frac{\pi}{4}$.

Integer powers of z

If $z = r(\cos \theta + i \sin \theta)$, then for $n = 0, 1, 2, \dots$, we have $z^n = r^n(\cos n\theta + i \sin n\theta)$.

Roots

If $z = w^n$ ($n = 1, 2, \dots$), then to each value of w there corresponds one value of z . We shall immediately see that conversely to a given $z \neq 0$, there correspond precisely n distinct values of w . Each of these values is called an n^{th} root of z and we write $w = \sqrt[n]{z}$. The n values of $\sqrt[n]{z}$ can be obtained as follows. We write z and w in polar form

$$\begin{aligned} z &= r(\cos \theta + i \sin \theta), \\ w &= R(\cos \phi + i \sin \phi). \end{aligned}$$

Then the equation $w^n = z$ becomes

$$w^n = R^n(\cos n\phi + i \sin n\phi) = z = r(\cos \theta + i \sin \theta).$$

The absolute values on both sides must be equal. Thus, $R^n = r$ so that $R = \sqrt[n]{r}$. Equating the arguments $n\phi$ and θ , we get $n\phi = \theta + 2k\pi$, where $k \in \mathbb{Z}$. Consequently, $\sqrt[n]{z}$, for $z \neq 0$ has the n distinct values

$$\sqrt[n]{z} = \sqrt[n]{r} \left(\cos \frac{\theta + 2k\pi}{n} + i \sin \frac{\theta + 2k\pi}{n} \right) \quad \text{for } k = 0, 1, 2, \dots, n-1.$$

These n values lie on a circle of radius $\sqrt[n]{r}$ with center at the origin and constitute the vertices of a regular polygon of n sides.

Derivative. Analytic Function

The circle of radius ρ and center a is given by the equation $|z - a| = \rho$. An open circular disk $|z - a| < \rho$ is called a ρ -neighborhood of a .

Half-planes: By the (open) upper half-plane we mean the set of all points $z = x + iy$ such that $y > 0$. Similarly, the condition $y < 0$ defines the lower half-plane, $x > 0$ the right half-plane and $x < 0$ the left half-plane.

Derivatives: The derivative of a complex function f at a point z_0 is written $f'(z_0)$ and is defined by

$$f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z},$$

provided this limit exists. Then f is said to be differentiable at z_0 .

A function $f(z)$ is said to be analytic in a domain D if $f(z)$ is defined and differentiable at all points of D . The function $f(z)$ is said to be analytic at a point $z = z_0$ in D if $f(z)$ is analytic in a neighborhood of z_0 .

Example

The nonnegative integer powers $1, z, z^2, \dots$ are analytic in the entire complex plane and so are polynomials, that is, functions of the form

$$f(z) = c_0 + c_1z + c_2z^2 + \dots + c_nz^n,$$

where $c_0, c_1, \dots, c_n$ are complex constants. The quotient of two polynomials $g(z)$ and $h(z)$, $f(z) = \frac{g(z)}{h(z)}$ is called a rational function. This f is analytic except at the points where $h(z) = 0$.

Cauchy-Riemann Equations. Laplace's Equation

The *Cauchy-Riemann equations* are the most important equations and one of the pillars on which complex analysis rests. They provide a criterion (a test) for the analyticity of a complex function

$$w = f(z) = u(x, y) + iv(x, y).$$

Roughly, f is analytic in a domain D if and only if the first partial derivatives of u and v satisfy the two Cauchy-Riemann equations

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

everywhere in D , here $u_x = \partial u / \partial x$ and $u_y = \partial u / \partial y$.

Let $f(z) = u(x, y) + iv(x, y)$ be defined and continuous in some neighborhood of a point $z = x + iy$ and differentiable at z itself. Then at that point, the first-order partial derivatives of u and v exist and satisfy the Cauchy-Riemann equations.

The great importance of complex analysis in engineering mathematics results mainly from the fact that both the real part and the imaginary part of an analytic function satisfy Laplace's equation, the most important PDE of physics which occurs in gravitation, electrostatics, fluid flow and heat conduction.

Laplace's Equation: If $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then both u and v satisfy Laplace's equation

$$\nabla^2 u = u_{xx} + u_{yy} = 0$$

and

$$\nabla^2 v = v_{xx} + v_{yy} = 0$$

in D and have continuous second partial derivatives in D .

Solutions of Laplace's equation having continuous second-order partial derivatives are called *harmonic functions* and their theory is called *potential theory*. If two harmonic functions u and v satisfy the Cauchy-Riemann equations in a domain D , they are the real and imaginary parts of an analytic function f in D . Then v is said to be a harmonic conjugate function of u in D .

Exponential Function

The complex *exponential function* e^z is defined by $e^z = e^x(\cos y + i \sin y)$. This definition is motivated by the fact the e^z extends the real exponential function e^x of calculus in a natural fashion.

Properties of exponential function:

- (i) e^z is analytic for all z .
- (ii) The derivative of e^z is e^z .
- (iii) $e^{z_1+z_2} = e^{z_1}e^{z_2}$.
- (iv) e^z is a periodic function with period $2\pi i$.

Trigonometric and Hyperbolic Functions. Euler's Formula

Trigonometric Functions: By using the Euler formulas, we can extend the familiar real trigonometric functions to complex trigonometric functions. The real cosine and sine functions are given by

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

and

$$\sin x = \frac{e^{ix} - e^{-ix}}{2}.$$

For complex value $z = x + iy$, complex trigonometric functions are defined as

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

and

$$\sin z = \frac{e^{iz} - e^{-iz}}{2}.$$

By the definition of complex trigonometric functions, it is clear that Euler's formula is valid in complex, i.e

$$e^{iz} = \cos z + i \sin z.$$

Another complex trigonometric functions are defined as

$$\tan z = \frac{\sin z}{\cos z}, \cot z = \frac{\cos z}{\sin z}, \sec z = \frac{1}{\cos z}, \csc z = \frac{1}{\sin z}.$$

Properties of trigonometric functions:

- (i) $\sin z$ and $\cos z$ are analytic functions for all z .
- (ii) $\sin z$ and $\cos z$ are periodic functions with period 2π .
- (iii) $\tan z$ and $\cot z$ are periodic functions with period π .

Hyperbolic Functions: The complex Hyperbolic sine and cosine functions are defined as

$$\cosh z = \frac{e^z + e^{-z}}{2}, \sinh z = \frac{e^z - e^{-z}}{2}.$$

Similarly other complex Hyperbolic functions are defined as

$$\tanh z = \frac{\sinh z}{\cosh z}, \coth z = \frac{\cosh z}{\sinh z}.$$

Complex trigonometric and Hyperbolic functions are related to each other by the following relation:

$$\cosh iz = \cos z, \quad \sinh iz = i \sin z$$

Relation between complex trigonometric and hyperbolic functions

$$\begin{aligned} \cosh iz &= \cos z, & \sinh iz &= i \sin z, \\ \cos iz &= \cosh z, & \sin iz &= i \sinh z. \end{aligned}$$

De Moivre's Theorem

If $z = r(\cos \theta + i \sin \theta)$, then for $n = 0, 1, 2, 3, \dots$, we have

$$z^n = r^n(\cos \theta + i \sin \theta)^n.$$

By using De Moivre's formula, we have

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$

Therefore,

$$z^n = r^n(\cos n\theta + i \sin n\theta).$$

Logarithm. General Power. Principal Value. Singularities and Zeros. Infinity

The *natural logarithm* of $z = x + iy$ is denoted by $\ln z$ (sometimes also by $\log z$) and is defined as the inverse of the exponential function; that is, $w = \ln z$ is defined for $z \neq 0$ by the relation

$$e^w = z.$$

If we set $w = u + iv$ and $z = re^{i\theta}$, this becomes

$$e^w = e^{u+iv} = re^{i\theta}$$

gives $e^u = r$ and $v = \theta$. We have

$$\ln z = \ln r + i\theta \quad (r = |z| > 0, \theta = \arg z).$$

Since the argument of z is determined only up to integer multiples of $2n\pi$, the complex natural logarithm $\ln z$ ($z \neq 0$) is infinitely many-valued.

The value of $\ln z$ corresponding to the principal value $\text{Arg } z$ is denoted by $\text{Ln } z$ and is called the principal value of $\ln z$. Thus,

$$\text{Ln } z = \ln |z| + i\text{Arg } z.$$

For every $n = 0, \pm 1, \pm 2, \dots$ the above formula defines a function, which is analytic except at 0 and on the negative real axis and has the derivative

$$(\ln z)' = \frac{1}{z}.$$

Each of the infinitely many functions is called a branch of the logarithm. The negative real axis is known as a branch cut. The branch for $n = 0$ is called the principal branch of $\ln z$.

General Powers: General powers of a complex number $z = x + iy$ are defined by the formula

$$z^c = e^{c \ln z} \quad (c \text{ complex, } z \neq 0).$$

Since $\ln z$ is infinitely many-valued, z^c will in general be multi-valued. The particular value $z^c = e^{c \text{Ln } z}$ is called the *principal value* of z^c .

Line Integral in The Complex Plane

Complex definite integrals are called (complex) *line integrals*. They are written

$$\int_C f(z) dz.$$

Here the integrand $f(z)$ is integrated over a given curve C or a portion of it. This curve C in the complex plane is called the *path* of integration. We may represent C by a parametric representation

$$z(t) = x(t) + iy(t) \quad (a \leq t \leq b).$$

C is said to be a smooth curve if C has a continuous and nonzero derivative at each point. Geometrically this means that C has everywhere a continuously turning tangent. All paths of integration for complex line integrals are assumed to be piecewise smooth, that is, they consist of finitely many smooth curves joined end to end.

Basic properties:

1. Linearity: Integration is a linear operation, that is,

$$\int_C [k_1 f_1(z) + k_2 f_2(z)] dz = k_1 \int_C f_1(z) dz + k_2 \int_C f_2(z) dz.$$

2. Sense reversal in integrating over the same path, from z_0 to Z (left) and from Z to z_0 (right), introduces a minus sign

$$\int_{z_0}^Z f(z) dz = - \int_Z^{z_0} f(z) dz.$$

3. Partitioning of path:

$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz.$$

Example

$$\int_0^{1+i} z^2 dz = \frac{z^3}{3} \Big|_0^{1+i} = \frac{(1+i)^3}{3} = -\frac{2}{3} + \frac{2}{3}i.$$

Bounds for Integrals: ML-inequality

$$\left| \int_C f(z) dz \right| \leq ML,$$

where L is the length of C and M is a constant such that $|f(z)| \leq M$ everywhere on C .

Example

Find an upper bound for the absolute value of the integral $\int_C z^2 dz$, where C is the straight line segment from 0 to $1 + i$.

Solution: The absolute value of the integral is

$$\left| -\frac{2}{3} + \frac{2}{3}i \right| = \frac{2}{3}\sqrt{2}.$$

Here, $L = \sqrt{2}$ and $|f(z)| = |z^2| \leq 2$ on C . Therefore,

$$\left| \int_C z^2 dz \right| \leq 2\sqrt{2}.$$

Cauchy's Integral Theorem

Cauchy's integral theorem is one of the fundamental results of complex analysis.

A *simple closed path* is a closed path that does not intersect or touch itself. For example, a circle is simple, but a curve shaped like an 8 is not simple.

A *simply connected domain* D in the complex plane is a domain such that every simple closed path in D encloses only points of D .

Examples: The interior of a circle, ellipse or any simple closed curve. A domain that is not simply connected is called multiply connected. Examples: an annulus or a disk without the center.

Cauchy's Integral Theorem: If $f(z)$ is analytic in a simply connected domain D then for every simple closed path C in D

$$\oint_C f(z) dz = 0.$$

Example

$$\oint_C e^z dz = 0, \quad \oint_C \cos z dz = 0, \quad \oint_C z^n dz = 0 \quad (n = 0, 1, 2, \dots)$$

for any closed path C , since these are entire functions.

Independence of path: We call an integral of $f(z)$ independent of path in a domain D if for every z_1, z_2 in D its value depends (besides on $f(z)$) only on the initial point z_1 and the terminal point z_2 , but not on the choice of the path C in D .

If $f(z)$ is analytic in a simply connected domain D , then the integral of $f(z)$ is independent of path in D .

As long as our deforming path always contains only points at which $f(z)$ is analytic, the integral retains the same value. This is called the *principle of deformation of path*.

Cauchy's Integral Formula

The most important consequence of Cauchy's integral theorem is Cauchy's integral formula. This formula is useful for evaluating integrals.

Cauchy's Integral Formula: Let $f(z)$ be analytic in a simply connected domain D . Then for any point z_0 in D and any simple closed path C in D that encloses z_0

$$\oint_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0)$$

the integration being taken counterclockwise.

Example

$$\oint_C \frac{e^z}{z-2} dz = 2\pi i e^z|_{z=2} = 2\pi i e^2$$

for any contour enclosing $z_0 = 2$ (since e^z is entire) and zero for any contour for which $z_0 = 2$ lies outside (by Cauchy's integral theorem).

Multiply connected domains If $f(z)$ is analytic on C_1 and C_2 and in the ring-shaped domain bounded by C_1 and C_2 and z_0 is any point in that domain, then

$$f(z_0) = \frac{1}{2\pi i} \oint_{C_1} \frac{f(z)}{z-z_0} dz + \frac{1}{2\pi i} \oint_{C_2} \frac{f(z)}{z-z_0} dz.$$

Derivatives of Analytic Functions

We use Cauchy's integral formula to show the basic fact that complex analytic functions have derivatives of all orders. Indeed, if a real function is once differentiable, nothing follows about the existence of second or higher derivatives. Thus, in this respect, complex analytic functions behave much more simply than real functions that are once differentiable.

If $f(z)$ is analytic in a domain D , then it has derivatives of all orders in D which are then also analytic functions in D . The values of these derivatives at a point z_0 in D are given by the formulas

$$f'(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz,$$

$$f''(z_0) = \frac{2!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^3} dz$$

and in general

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz \quad (n = 0, 1, 2, \dots).$$

Here C is any simple closed path in D that encloses z_0 and whose full interior belongs to D and we integrate counterclockwise around C .

Cauchy's Inequality: If $|f(z)| \leq M$, then

$$|f^{(n)}(z_0)| \leq \frac{n!M}{r^n}.$$

Liouville's Theorem: If an entire function is bounded in absolute value in the whole complex plane, then this function must be a constant.

Morera's Theorem: If $f(z)$ is continuous in a simply connected domain D and if

$$\oint_C f(z)dz = 0$$

for every closed path in D , then $f(z)$ is analytic in D .

Taylor and Maclaurin Series

The *Taylor series* of a function $f(z)$, the complex analog of the real Taylor series is

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n,$$

where $a_n = \frac{1}{n!} f^{(n)}(z_0)$ or $a_n = \frac{1}{2\pi i} \oint_C \frac{f(z^*)}{(z^* - z_0)^{n+1}} dz^*$. Here, we integrate counterclockwise around a simple closed path C that contains z_0 in its interior and is such that $f(z)$ is analytic in a domain containing C and every point inside C . The remainder of the Taylor series after the term $a_n(z - z_0)^n$ is

$$R_n(z) = \frac{(z - z_0)^{n+1}}{2\pi i} \oint_C \frac{f(z^*)}{(z^* - z_0)^{n+1}(z^* - z)} dz^*.$$

Thus, we have

$$f(z) = f(z_0) + \frac{(z - z_0)}{1!} f'(z_0) + \frac{(z - z_0)^2}{2!} f''(z_0) \\ + \dots + \frac{(z - z_0)^n}{n!} f^{(n)}(z_0) + R_n(z).$$

This is called Taylor's formula with remainder. A *Maclaurin series* is a Taylor series with center $z_0 = 0$.

Taylor's theorem: Let $f(z)$ be analytic in a domain D and let $z = z_0$ be any point in D . Then there exists precisely one Taylor series with center z_0 that represents $f(z)$. This representation is valid in the largest open disk with center z_0 in which $f(z)$ is analytic. The remainders $R_n(z)$ can be represented in the form defined above. The coefficients satisfy the inequality

$$|a_n| \leq \frac{M}{r^n},$$

where M is the maximum of $|f(z)|$ on a circle $|z - z_0| = r$ in D whose interior is also in D .

Exponential Function: It is known that the exponential function e^z is analytic for all z and $(e^z)' = e^z$. Therefore, we obtain Maclaurin series

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$$

Trigonometric Functions:

$$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots$$

$$\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$$

Hyperbolic Functions:

$$\cosh z = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} = 1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \dots$$

$$\sinh z = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} = z + \frac{z^3}{3!} + \frac{z^5}{5!} + \dots$$

Logarithm Function:

$$\ln(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \dots$$

1. Advanced Engineering Mathematics by Erwin Kreyszig, John Wiley, 10th edition, 2011.